Microbotox of the Lower Face and Neck: Evolution of a Personal Technique and Its Clinical Effects

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Summary: Microbotox is the injection of multiple microdroplets of diluted onabotulinumtoxinA into the dermis or the interface between the dermis and the superficial layer of facial muscles. The intention is to decrease sweat and sebaceous gland activity to improve skin texture and sheen and to target the superficial layer of muscles that find attachment to the undersurface of the dermis causing visible rhytides. For treatment of the lower face and neck, hundreds of microdroplets of diluted Botox are injected into the dermis or immediate subdermal plane to improve skin texture, smoothen horizontal creases, and decrease vertical banding of the neck, as well as to achieve better apposition of the platysma to the jawline and neck, improving contouring of the cervicomental angle. The Microbotox solution is mixed in the syringe by adding a small volume of lidocaine to the calculated dose of onabotulinumtoxinA drawn from a standard bottle of Botox prepared with 2.5mL saline. Each 1mL syringe of Microbotox solution contains 20–28 units of onabotulinumtoxinA per mL of solution and is used to deliver 100–120 injections. The lower face and neck will usually require 1mL per side. The injections are delivered intradermally using a 30- or 32-G needle raising a tiny blanched weal at each point. The author has over 1867 documented cases of Microbotox in various parts of the face (forehead, glabellar, crow's-feet, infraorbital, and cheeks) and neck, the majority of these patients being treated in forehead or the lower face and neck as described in this article. (*Plast. Reconstr. Surg.* 136: 92S, 2015.)

Microbotox, which has been previously described,[1–3] is a term coined in 2001 to describe the author's technique (developed in 2000) of delivering multiple intradermal or subdermal injections of diluted onabotulinumtoxinA into the overlying skin envelope and superficial layer of the facial and neck muscles. The aim of this technique is to specifically target the sweat and sebaceous glands in the skin and the superficial fibers of the facial muscles that take attachment to the undersurface of the dermis. This technique is different in concept and approach compared with other intradermal techniques that have been published. Initially, the technique was called Mesobotox as the method of delivery seemed similar to the technique of Mesotherapy, but it was later felt that this term was neither sufficiently scientific nor did it convey the extent or depth of the injections over the target areas. The term "Microbotox" was chosen instead to reflect the small doses of toxin delivered in each microdroplet. The Microbotox technique was developed by the author (W.T.L.W.) using onabotulinum toxin type A (Botox; Allergan Pharmaceuticals Ireland, Westport, County Mayo, Ireland) primarily, and for this reason, the toxin will be referred to in the text as Botox or Microbotox.

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Microbotox refers to the systematic injection of multiple tiny blebs of diluted Botox at 0.8- to 1.0-cm intervals into the skin or just below that into the superficial fibers of the facial muscles. The intent is not to completely paralyze the underlying facial muscles but only to weaken the superficial fibers that are inserted into the undersurface of the skin, which are responsible for the fine lines and wrinkles on the face and neck. These intradermal injections would have the desired effect of smoothening and tightening the skin (due to the neurochemically induced bulk atrophy of the sweat and sebaceous glands),[4–8] as well as by decreasing the subtle pulling and tethering effects of the facial muscles where they are attached to the skin. Small injection volumes would prevent unwanted diffusion of the solution into deeper muscles, thus retaining more muscle function to give a natural appearance. This is especially useful in the forehead or under-eye regions where traditional Botox dosage and droplet size often leads to a stiff immovable forehead or inanimate lower eyelids.

HISTORICAL CONTEXT

The Microbotox technique was developed in response to the way Botox had traditionally been administered since 1995 when the author first started using it. Back then, it was recommended to use a dilution of 2.0–2.5 mL (the most common dilution) of saline for every 100 units of Botox. Injections of 2–4 Botox units (0.05–0.1 mL) were delivered to target muscles, such as the corrugators, depressor supercilii, procerus, superior orbicularis, lateral orbicularis, and the frontalis. The paralysis of these muscles was usually complete, and it was initially gratifying to be able to achieve a smooth brow, forehead, and lateral orbital region with only a few droplets of a purified protein complex. However, after the euphoric satisfaction in the nineties of being able to induce the paralysis of muscles and reduce the appearance of periorbital wrinkles by mere injections alone, we encountered many patients who complained of a frozen forehead, stiff immovable brows, and an unnatural, inanimate appearance on smiling. They wanted to have more natural and less dramatic results. We started to experiment with reducing the dose, increasing the dilution, spreading out the injections, or injecting only specific muscles in a limited fashion to refine the results.

Over 13 years ago, the author described in various scientific meetings his Botox brow lifting and brow shaping techniques using traditional doses of 2–4 units (0.05–0.1 mL) per injection point and the technique of Botox facial slimming that reduced lower facial width by decreasing the volume and activity of the masseter muscles and the parotid glands.[9–11] At the same time, it was observed that if the Botox was lightly injected (small blebs, low doses) into the dermis or the interface between dermis and the superficial facial muscles, the appearance of the wrinkles and fine lines would be decreased and yet sufficient movement of the underlying muscles would be retained to confer a more natural appearance to the face in the injected region. In addition, the smooth lustrous appearance of the forehead skin due to decreased sweat and sebaceous gland activity was an aesthetic bonus. This latter was an effect that had already been noticed whenever intramuscular injections of Botox into the forehead were delivered, as the blebs used then were so large that, inevitably, some Botox diffused into the dermis creating an unintended intradermal effect. The objective of Microbotox was therefore to deliver the tiny blebs of Botox as superficially as possible, specifically to create this intradermal effect and superficial muscle weakening and at the same time not to allow diffusion of the toxin into the entire muscle to paralyze it completely. A comparison between the mechanism of action of standard Botox versus Microbotox is shown in Figure 1.

The effects in the brow and forehead were encouraging with many patients repeatedly requesting for Microbotox treatments so as to have an attenuation of disturbing lines and wrinkles yet to retain a natural animated appearance. The use of Microbotox was then extended to the under-eye lines, to decrease oiliness of the central T zone of the face and cheeks, and eventually to the lower face and neck.

MICROBOTOX OF THE LOWER FACE AND NECK

The area of the lower face and neck that benefits from Microbotox injections is defined as a zone bounded by a line 3 fingerbreadths above and parallel to the lower border of the mandible, 1 fingerbreadth posterior to the depressor anguli oris, and all the way down over the cervicomental angle and anterior neck to the upper border of the clavicle, and posteriorly to the anterior border of the sternocleidomastoid. This corresponds with the anatomical extent of the platysma as it sweeps up from the clavicle over the jawline to blend in with the superficial musculoaponeurotic system

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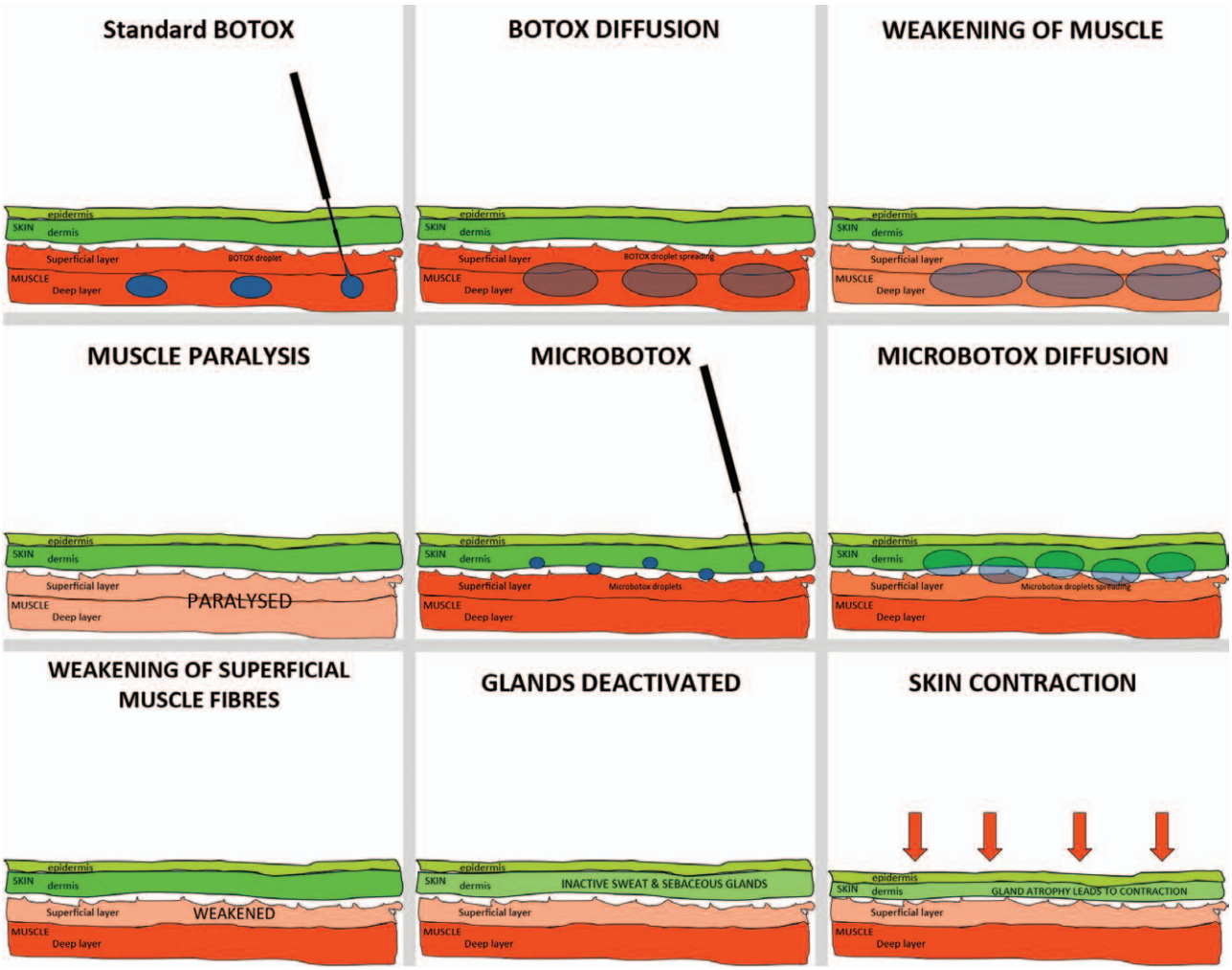


Fig. 1. A comparison between the concept of standard Botox and Microbotox droplet size, placement, diffusion, and effects. (*Above, left*) Standard Botox is delivered into the muscles as droplets of volume 0.05–1.0 mL (2–4 units) using a dilution of 2.5 mL to 100 units of Botox. (*Above, center*) The Botox diffuses through both superficial and deep fibers of the muscle. (*Above, right*) The entire muscle gets weakened. (*Center, left*) The entire muscle is paralyzed. (*Center, center*) Microbotox droplets are very small and delivered to the dermis or the junction of the dermis with the superficial fibers of the facial muscles. This is where the muscle fibers are attached to the undersurface of the dermis. (*Center, right*) The Microbotox droplets can only diffuse into the dermis and the superficial layer of the muscle. (*Below, left*) Only the superficial muscle fibers are weakened. (*Below, center*) The sweat and sebaceous glands are deactivated due to the effect of the Microbotox. (*Below, right*) Bulk atrophy of the sweat and sebaceous glands leads to a decrease in volume and thickness of the dermal layer, which brings about a subtle contraction of the overlying skin envelope and a sensation of "tightening."

of the face (Fig. 2). Injections are delivered only outside of this area over and posterior to the sternocleidomastoid if there are horizontal neck lines that extend far laterally or if there are particularly strong lateral platysmal bands. The intent of delivering Microbotox into this area is not only to improve neck skin texture, color, and smoothness but also to decrease the activity of the superficial fibers of the platysma muscle that in turn creates a lifting effect of the jowls and jawline, as well as better cervicomental contouring.

INDICATIONS

Microbotox of the lower face and neck is a simple nonsurgical solution that is ideal for patients who seek improvement of mild neck laxity and jowling, crepey neck skin with a rough texture, and horizontal neck lines and vertical banding, but do not wish to undergo surgery. Those are patients who are already receiving other nonsurgical treatments to counter the signs of aging. It is best indicated when there are early signs of lower face and neck aging that

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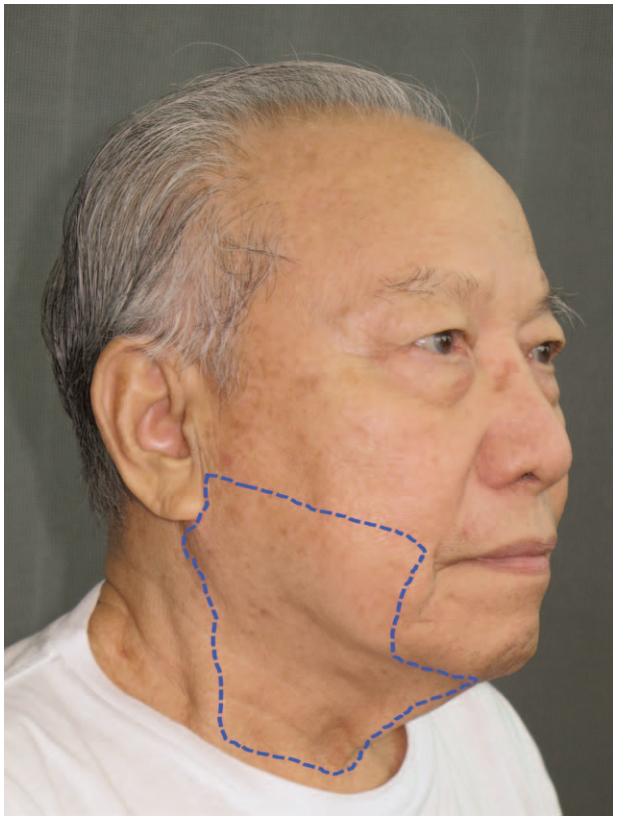


Fig. 2. The margins of the area to be injected in Microbotox of the lower face and neck—delineated by a continuous line 3 fingerbreadths above the mandibular border, posterior to the depressor anguli oris, over the entire neck and cervicomental region anteriorly, and limited posteriorly by the border of the sternocleidomastoid. This corresponds to the extent of the platysma muscle.

may not warrant the invasiveness and extent of a surgical face-lift. It can be used in conjunction with other skin tightening devices and therapies. Patients should be made aware that the effects do not last more than several months and that the procedure needs to be repeated periodically to maintain the result. Loss of result is therefore not considered a failure.

THEORETICAL MECHANISM OF ACTION IN THE PLATYSMA

Consider any muscle as a 3-dimensional tubular structure with a defined volume. When it contracts, it shortens and the diameter or girth of the tube increases. The vertical contraction creates a lateral force or shift (bulging) on either side of the muscle. If only one side of the cylindrical muscle is paralyzed, then that side will not contract but the other side will and it will exert a pulling action in the opposite direction.

The platysma muscle is a large sheet of subcutaneous muscle belonging to a group of unique muscles called the panniculus carnosus. These muscles lie in a plane just under the panniculus adiposus. Other muscles that belong to this group are the palmaris brevis in the hand, the dartos muscle in the scrotum, and the corrugator cutis ani. In the case of the platysma muscle, when it contracts it shortens the distance between the chin and the clavicle with blunting of the cervicomental angle (Fig. 3, *left* and *center*). This can be easily seen by contracting the muscle. Weakening the superficial fibers of the platysma muscle with Microbotox injections decreases the appearance of horizontal neck lines and vertical bands, while the deeper muscle fibers continue to function and exert their muscle pull inward and upward creating a more snug apposition of the platysma to the underlying neck structures. This creates a more-defined cervicomental angle and the illusion of lifting of the jowls and jawline (Fig. 3, *right*). This theoretical mechanism of action is called the "platysma effect." The smoothening of the neck skin due to the intradermal effects as already described completes the effect of an improved lower face and neck appearance.

HOW TO PREPARE THE MICROBOTOX

For most parts of the face such as the glabellar, crow's-feet, forehead, and infraorbital regions, the concentration of Microbotox used is 20 units in 1 mL of solution. This is equivalent to reconstituting a bottle of Botox with 5.0 mL saline and then directly drawing out 1 mL of the solution into the syringe. However, it is more convenient to use Botox from a bottle of 100 units that has been reconstituted with 2.5 mL of saline (standard dilution). Further dilution to the appropriate Microbotox concentration is then done in the syringe itself. This allows the bottle of standard Botox to be retained should it be needed in a more concentrated form for masseteric injections or when needed to administer more concentrated Botox to key points of the face. This decreases the likelihood of delivering a droplet volume that is too large, which could diffuse out of the target zone and create unintended effects. It also allows the concentration of the Microbotox solution to be varied. In patients with thin necks, a dilution of 20 units per mL solution is sufficient. In patients with visibly thicker necks or deep horizontal neck lines, a concentration of 28 units per mL solution gives better results.

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Fig. 3. (*Left*) The platysma muscle has both a deep layer and superficial layer of muscles that attach at multiple points into the undersurface of the neck skin. The platysma at rest adorns the contour of the neck. With aging and passive shortening of the muscle, vertical bands may appear. (*Center*) During active contraction of the platysma, there is a shortening of the muscle and a blunting of the cervicomental angle. There is also a concomitant elevation of the skin over the upper chest and a depression of the jowls. (*Right*) The platysma effect—if the superficial muscle fibers that are attached to the skin are weakened with Microbotox, the horizontal lines are reduced and the remaining (deeper) active platysma contracts inward creating a more snug apposition of the platysma-skin layer to the neck and jawline.

To prepare the Microbotox solution, 0.5 mL (20 units) from a standard bottle is drawn into a 1 mL syringe. A further 0.5 mL of lidocaine (0.5%) with or without adrenaline is then drawn into the syringe to make it up to 1 mL volume. This gives a concentration of 20 units Botox per mL. If 28 units is required, then 0.7 mL (28 units) is drawn from a standard bottle and then topped up with an additional 0.3 mL lidocaine (0.5%). The syringe is then ready for injection using a 30- or 32-G needle with the bevel pointed downward.

TECHNIQUE

In the average lower face and neck, 3 syringes of 20 units per mL Microbotox are used to cover the entire area as previously described. In heavier necks, 3 syringes of 28 units per mL Microbotox are used. With good injection technique, sometimes only 2 syringes are used. This depends on the skill of the injector and accurate delivery of the correct size droplet into the dermis. Quite often, there is droplet loss through poor technique or the Microbotox is delivered too deeply and too much squirted in one place. It is important to ensure that no air bubbles are present in the syringe as this will affect accurate delivery of the droplets.

The patient is first prepared with a layer of Elamax (5%) (Ela-Max5; Ferndale Laboratories, Inc., Ferndale, MI) over the area to be injected. After 20 minutes, this is thoroughly washed off.

The patient is then placed in a semi-reclined position with the neck arched upward to make the skin taut. Additional finger pressure helps to maintain this tautness when the injections are given. Injection technique is important. It is useful to practice delivering these microdroplets consistently (Fig. 4). Care must be taken to just penetrate the needle into the skin as superficially as possible. A resistance should be felt on pressing the plunger, and a small raised, blanched bleb in the skin should be seen. This indicates good depth of injection. If the solution is easily injected, the needle has probably been inserted too deeply and is subdermal or intramuscular. Despite over 200 injections being delivered, patients are comfortable and tolerant of the

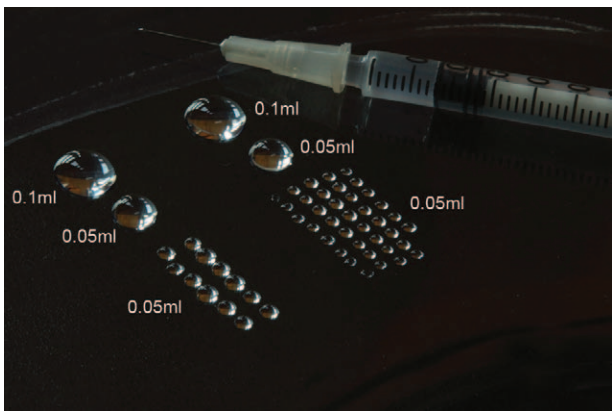


Fig. 4. It is useful to practice delivering the microdroplets consistently. Using a standard 2.5-mL dilution of 100-unit Botox, a 0.1-mL droplet contains 4 units. A 0.05-mL droplet contains 2 units. The Microbotox droplet is very tiny as shown in the diagram, and a 0.05-mL droplet can further be divided into 20–30 smaller Microbotox droplets each bearing a small dose of Botox.

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procedure and voluntarily returning for repeat treatments. This is largely due to the lidocaine which has been mixed into the Microbotox solution. The initial prick may be painful, but within 1 or 2 seconds, the injection point becomes anesthetized and no longer bothers the patients as the rest of the injections are completed in remaining areas. At the end of the procedure, patients do not experience any pain (Fig. 5).

As a general gauge of competence and proficiency with the technique, an injector should be able to deliver between 100 and 120 injections with each 1 mL syringe of Microbotox solution. (**See Video, Supplemental Digital Content 1**, which demonstrates Dr. Wu's Microbotox injection technique for the lower face and neck, available in the "Related Videos" section of the full-text article on PRSJournal.com or, for Ovid users, at ***http://links.lww.com/PRS/B433***).

RESULTS

Clinically, the cervicomental angle and jawline appear sharper and the jowls appear lifted as the platysma is allowed to conform more closely to the underlying neck shape. This is the platysma effect (Fig. 6). Neck bands are reduced and the skin appears smoother and firmer. Disturbing crepiness, skin bunching, and creasing, which happen on contraction of the platysma in middle-aged patients to older patients, are significantly reduced giving the appearance of a smoother and cleaner neck skin (Fig. 7). Patients express a feeling of desired tightness of the neck and jawline. This is helpful in patients who have undergone reduction of masseteric and/or parotid gland hypertrophy where the overlying skin envelope can go through a period of initial skin laxity and visible sagging. The Microbotox helps to tighten the skin envelope through the bulk atrophy of sweat and sebaceous glands as well by directing the force of platysmal pull inward to the cervicomental angle.

The duration of improvement after a session of Microbotox typically lasts for 3–4 months and in some cases up to 6 months.

Since 2000 when the technique was first developed, the author has treated over 2000 patients with Microbotox. Patients treated in the first year were not clearly documented; but subsequently, 1867 of these patients have been recorded and documented and a retrospective analysis is pending. These patients received Microbotox in various parts of the face (forehead, glabellar, crow's-feet, infraorbital, and cheeks) and neck, with the majority of these patients being treated in the forehead or in the lower face and neck as described in this article. Many patients had Microbotox administered to multiple regions at the same time thus increasing the total number of procedures documented.



Fig. 5. A 60-year-old woman seen in the anterior and oblique views having received Microbotox of the lower face and neck and showing the number of injection points that correspond to the distribution of the platysma muscle as it flows up the neck over the jawline and into the lower third of the face. Care is taken to avoid injecting over the depressor anguli oris in case of an asymmetric smile and over the sternocleidomastoid in case of unintended neck weakness. Two syringes of Botox 20 units in each mL were used to cover the entire neck. A total of 40 units was used. Botox injections of the chin are usually given in the usual standard fashion using a 2.5-mL dilution.

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Video. Supplemental Digital Content 1, demonstrating Dr. Wu's Microbotox injection technique for the lower face and neck, is available in the "Related Videos" section of the full-text article on PRSJournal.com or, for Ovid users, at ***http://links.lww.com/prs/b433***.

COMPLICATIONS

Most complications have arisen from mistakes with droplet size and depth of injection. If the injections are delivered subdermally and the droplet size is larger than what has been recommended, the Microbotox will diffuse into the thickness of the underlying muscle creating total paralysis rather than mere superficial muscle weakening. Similarly, if the droplet has been correctly delivered intradermally but is too large, it will also diffuse more deeply into the underlying muscle. When this happens, patients have complained of a stiff immovable brow when delivered to the forehead region and weakness of the sternocleidomastoid muscles and depressor anguli oris when delivered to the lower face and neck. Asymmetry and a lopsided smile have been seen in some



Fig. 6. A 65-year-old man with complaints of a heavy neck and sagging jawline but not wishing to undergo surgery or receive any skin tightening treatments. For over 5 years, he has been satisfied with the Microbotox treatment, which is administered every 6 months. Patients with heavier necks and visibly thicker platysmal mass such as this man require a higher dose than a woman with a thin neck. He received 3 syringes of 28 Botox units in 1-mL solution. A total of 84 units was used. (*Above, left*) Preinjection right oblique view. (*Above, right*) Two weeks postinjection right oblique view. (*Below, left*) Preinjection side view. (*Below, right*) Two weeks postinjection side view.

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Fig. 7. A 56-year-old woman with complaints of early neck laxity, deep horizontal neck lines, and crepey, finely wrinkled neck skin. Three syringes of 20 Botox units in 1 mL each were used; a total of 60 Botox units. Two weeks post injection, the skin appears smoother and cleaner with better light reflectance. The horizontal neck lines and vertical banding are reduced and the patient experiences better cervicomental contouring. (*Above, left*) Front view of patient seen before Microbotox. (*Above, right*) Front view of patient 2 weeks after Microbotox neck and lower face treatment. (*Below, left*) Preinjection right oblique view. (*Below, right*) Two weeks after Microbotox injection.

patients. These complications spontaneously subside within 2–3 weeks. There have been no cases of dysphonia, swallowing difficulties, or dry mouth.

DISCUSSION

The Microbotox technique is easy to learn and administer. On a few occasions, the author has employed the Microbotox technique using incobotulinumtoxinA (XEOMIN; Merz Pharmaceuticals, Greensboro, NC) and botulinum toxin type A products from China (CBTX-A; Lanzhou Biological Products Institute, Lanzhaou, China) and Korea (Medytox, Seoul, South Korea). The dilution protocols and effects of these products were observed to be similar to those of onabotulinumtoxinA. The author has also tried the technique using abobotulinumtoxinA (Dysport; Ipsen SA, Paris, France), but it was more difficult to predict the outcome. In some cases using the abobotulinumtoxinA, the muscle paralysis was too profound. In other cases, the effect was insufficient. This may be due to the lack of a fixed dose ratio between Dysport and Botox. However, familiarity with the varying degrees of dilution of these different toxins and careful observation and documentation of their results should lead to more predictable outcomes and allow this "Microtoxin" technique to be employed using any of the toxins readily available.

In preparing the Microbotox solution with any of these toxins, the use of lidocaine has been

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useful in decreasing the lingering sensation of pain and soreness after each injection. Considering that a typical treatment session of lower face and neck might require over 200 injections, any trick to aid in the reduction of pain is not insignificant. Elamax (5%) can also be used to decrease the sensation of pain, but in our experience, patients benefit more from the addition of lidocaine to the final solution. That is to say that patients who have had only Elamax followed by Microbotox injections without lidocaine experience more intraprocedural and postinjection discomfort than those who have not had Elamax but the Microbotox solution containing lidocaine.

Although there may be some concerns that lidocaine may diminish the full effect or duration of the toxin, the author has not noticed any of these changes. Perhaps, this is because the syringes of Microbotox diluted with lidocaine are administered immediately. Similarly, any bottles of standard Botox mixed with lidocaine are used within a few hours and never left to store overnight.

Lidocaine with adrenaline can be used in the Microbotox solution, but patients experience more sting at the time of injection. It also leaves multiple points of blanching, which is useful for teaching purposes, but bona fide patients will dislike this temporary speckled appearance. In clinical practice, the use of plain lidocaine is preferred.

Microbotox has been found to be a useful adjunct in patients undergoing Botox injections to the masseters and/or parotid glands to decrease lower facial width. These latter patients often experience a temporary jowling and blunting of the jawline in the first month after commencing treatment. This may be due to the inability of the overlying skin envelope to keep pace and contract as quickly as the masseter/parotids are devolumizing. Microbotox of the lower face and neck has also been used routinely in patients undergoing skin tightening treatments with heat delivery devices.

CONCLUSIONS

Microbotox of the lower face and neck is a useful nonsurgical adjunct to improve the neck and jawline contour and to improve the texture, color, and sheen of the neck skin and horizontal lines and vertical banding. It can be used in conjunction with other skin tightening devices and the use of fillers to revolumize the pre- and post-jowl sulci. It is particularly useful in patients who concomitantly undergo Botox facial slimming (masseteric and parotid gland reduction).

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PATIENT CONSENT

Patients provided written consent for the use of their images.

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